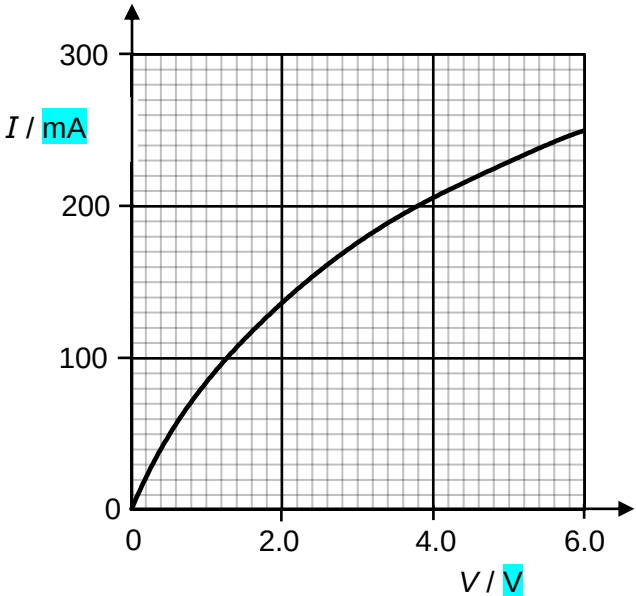


7	(a)	Fig. 7.1 shows the variation with potential difference V of the current I for a filament lamp X rated at 6.0 V and 1.5 W.  Fig. 7.1		
	(i)	Calculate the resistance of the filament lamp at 6.0 V.		
		resistance = Ω		
				[2]
	L1	<i>R is given by the RATIO of the coordinates (V, I) at the operating point of interest.</i> $R = \frac{V}{I} = \frac{6.0}{250 \times 10^{-3}}$		M1 A1

[Turn over

			$= 24 \Omega$	
	(ii)	Explain how Fig. 7.1 shows that the resistance of the filament lamp increases as the potential difference across the filament lamp increases.		
				
				[1]
	L1	Fig. 7.1 shows that as V increases, the <u>ratio of V to I</u> increases. Since resistance is defined as the <u>ratio</u> of potential difference V to the current I, the resistance of the filament lamp increases as the p.d. increases.		B1
	(iii)	The filament lamp X is connected in series with a battery B and a variable resistor R, and, in parallel with a voltmeter, as shown in Fig. 7.2. Fig. 7.2 Battery B has electromotive force 6.0 V and internal resistance 0.25 Ω. The resistance of R is adjusted such that the voltmeter reads 5.0 V. Calculate the terminal potential difference (p.d.) across B. Show your working.		
		terminal p.d. across B = V		[3]
	L2	From Fig. 7.1, when $V = 5.0 \text{ V}$ across lamp X, current through lamp X is $I = 230 \text{ mA} = 0.230 \text{ A}$. <i>*Note: Resistance of X is <u>not the same</u> as that in (a)(i). Do <u>not</u> accept $I = 5.0 \text{ V} / 24 \Omega = 0.208 \text{ A}$.</i> Terminal p.d. across B, V_t $= E - Ir$ $= 6.0 - (0.230)(0.25)$ $= 5.9425 = 5.9 \text{ V}$		M1 M1 A1

**Note: Both M1 and A1 marks are not awarded if the wrong value of current I is substituted or if working is not shown clearly.*

(b) Fig. 7.3(a) and Fig. 7.3(b) show two circuits which can be used to act as a **dimmer switch** for a lamp.

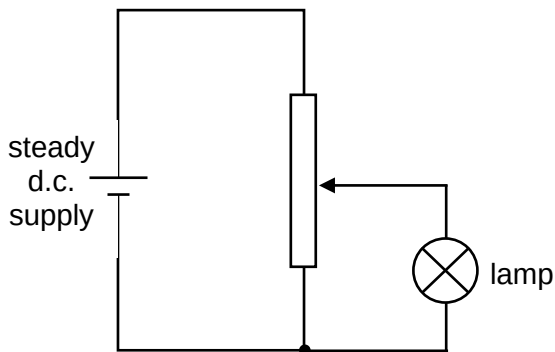


Fig. 7.3(a)

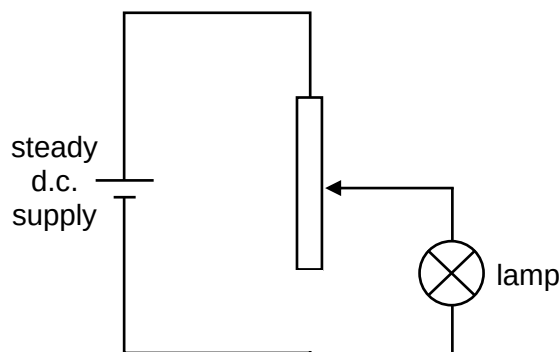


Fig. 7.3(b)

State and explain **one advantage** the circuit in Fig. 7.3(a) has over the circuit in Fig. 7.3(b) in **varying the brightness** of the lamp.

.....

.....

[2]

L3 State: **The lamp in circuit in Fig. 7.3(a) can be fully switched off.**
 Explain: **In the circuit in Fig. 7.3(b), there is always p.d. across the lamp.**

Do not accept "waste less energy/power". Answer too generic and there is still current flowing through the fixed resistor hence there is still power dissipated in the resistor even when the lamp is off in Fig. 7.3(a).

B1
B1

[Total: 8]